Answer Key & Teacher Notes

Session 1 — Median and Outliers · Session 2 — Comparing Distributions

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Median and Outliers

TALK PROMPTS

- Why does the 120-car weekend have less effect on the median than on the mean?
- Can an outlier be very low? What happens to the median then?

PARTNER TASK

Field Goal Kicks. One partner orders 33, 36, 35, 38, 34, 37, 72 and finds the median; the other names the outlier and predicts what happens to the median if 72 becomes 92. Swap with Test Corrections.

WATCH FOR

- Calling the outlier the typical value because it stands out.
- Thinking a bigger outlier always makes the median bigger.
- Forgetting that an outlier can be very low as well as very high.

SESSION 2 · TEACHER NOTES

Comparing Distributions

TALK PROMPTS

- How can a lower median wait and a smaller IQR both point to the same line?
- When would you choose the group with the lower median?

PARTNER TASK

Two Carrot Plots. One partner finds the median and IQR for Plot A; the other for Plot B. Together write one sentence about center and one about spread.

WATCH FOR

- Thinking the group with the higher median is also the more consistent one.
- Comparing only medians and ignoring the IQRs.
- Treating a smaller IQR as a lower center.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.5d	1	1	2	2	2	2	3	3	2	1	1
S1 · Form B	6.SP.B.5d	1	1	2	2	2	2	3	3	2	1	1
S2 · Form A	6.SP.B.5d	2	1	2	3	2	3	3	3	2	1	1
S2 · Form B	6.SP.B.5d	2	1	2	3	2	3	3	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

outlier: A value **far** from all the others.

median: The **middle** value of the ordered data.

mean: The sum of the values **divided** by how many values there are.

- An outlier pulls the **mean** toward it.
- The median stays in the **middle**, so it describes typical data better when there is an outlier.

Example: Order: 39, 41, 43, 44, 45, 47, 120. The outlier is **120**. The median is the 4th value: **44**. Answer: A typical Saturday has about **44** cars.

YOU TRY

1. Order: 11, 12, 13, 14, 15, 16, 60. a. 60 b. 14

2. 14 (the middle value does not change)

EXIT TICKET

1. Order: 3, 20, 21, 22, 23, 24, 25. a. 3 b. 22

2. The median, $5. The $40 outlier pulls the mean up to $12, more than 4 of the 5 prices.

3. **B** (more than the median)

PRACTICE FORM A

1. Order: 39, 41, 43, 44, 45, 47, 120. a. 120 b. 44 c. the median, 44; 120 is far from the rest

2. Order: 33, 34, 35, 36, 37, 38, 72. a. 72 b. 36

3. Order: 4, 5, 6, 7, 8, 9, 52. a. 7 b. 7 (no change)

4. Order: 55, 57, 58, 60, 62, 64, 210. Median = 60 dollars. Six of the seven prices are close to 60; 210 is an outlier, not typical.

5. It stays the same (48 min). The largest value stays at the top of the order, so the middle value does not change.

6. Order: 5, 82, 84, 86, 88, 90, 92. a. 86 b. 86 (no change)

7. a. 24 (order 18, 20, 22, 24, 25, 27, 96) b. 96 c. The median, $24. The outlier $96 pulls the mean up, so the mean is greater than 6 of the 7 days.

8. a. $30,500 (order: the middle two are $30,000 and $31,000) b. The $210,000 salary pulls the mean up; 5 of 6 workers earn far less than $60,000. c. The median, because it describes what a typical worker earns.

9. Standing out is not the same as typical. 250 is an outlier. The median, 47, describes a typical value.

10. **B** (an outlier)

11. **A** (7)

Self-check word **OUTLIERS**

PRACTICE FORM B

1. Order: 48, 50, 51, 52, 53, 55, 130. a. 130 b. 52 c. the median, 52; 130 is far from the rest

2. Order: 41, 42, 43, 44, 45, 46, 85. a. 85 b. 44

3. Order: 5, 6, 7, 8, 9, 10, 48. a. 8 b. 8 (no change)

4. Order: 24, 26, 28, 29, 31, 33, 140. Median = 29 dollars. Six of the seven prices are close to 29; 140 is an outlier, not typical.

5. It stays the same (35 min). The largest value stays at the top of the order, so the middle value does not change.

6. Order: 3, 74, 77, 79, 81, 83, 85. a. 79 b. 79 (no change)

7. a. 33 (order 27, 29, 31, 33, 34, 36, 110) b. 110 c. The median, $33. The outlier $110 pulls the mean up, so the mean is greater than 6 of the 7 days.

8. a. $35,500 (order: the middle two are $35,000 and $36,000) b. The $244,000 salary pulls the mean up; 5 of 6 workers earn far less than $70,000. c. The median, because it describes what a typical worker earns.

9. Standing out is not the same as typical. 200 is an outlier. The median, 38, describes a typical value.

10. **B** (median)

11. **B** (up)

Self-check word **STANDOUT**

GUIDED NOTES

distribution: How the data values are **spread** across their range.

center (median): The **typical** value of a data set.

spread (IQR): How **close together** the middle half is.

- Use the **median** to compare typical values and the IQR to compare consistency.
- A higher median does not mean **steadier**; only the IQR shows that.

Example: Center: Line 1's typical wait is **5** min shorter. Spread: Line 1's IQR is **smaller**, so its waits are steadier. Answer: Join Line **1**.

YOU TRY

1. a. Group 2 b. Group 1

2. Same spread (IQR 8); Team B's typical value is 14 higher.

EXIT TICKET

1. a. Shop K b. same (both IQR 8)

2. 8; 12 − 5 = 7

3. **C** (E is higher; F is more consistent)

PRACTICE FORM A

1. Line 1: median 8, Q1 5, Q3 12, IQR 7. Line 2: median 13, Q1 4, Q3 16, IQR 12. Join Line 1: shorter typical wait and steadier.

2. a. 15 b. 17 − 12 = 5 c. 15 d. 23 − 11 = 12. Same center; Plot A is more consistent.

3. a. Month 2 b. Month 2

4. Not right. E has the higher center (60 > 55), but F is more consistent (IQR 8 < 30).

5. a. 20 b. 8 c. 20 d. 16. e. Same median; Cooler 1 is more consistent (smaller IQR).

6. Team 2. Its IQR is much smaller (5 vs. 20), so its scores stay close to its median.

7. Store Q usually sells more (median 52 vs. 46), but Store P is more consistent (IQR 9 vs. 21).

8. a. Method 2 b. Method 1 (IQR 6 vs. 24) c. Model: Method 1; its typical score is only 5 points lower, and almost every student scores close to it.

9. Median measures center and IQR measures spread; one does not decide the other. Example: Set E has median 60, IQR 30 and Set F has median 55, IQR 8. Compare each measure separately.

10. **A** (median and IQR)

11. **B** (more consistent)

Self-check word **CENTERS**

PRACTICE FORM B

1. Booth 1: median 9, Q1 6, Q3 13, IQR 7. Booth 2: median 14, Q1 3, Q3 17, IQR 14. Join Booth 1: shorter typical wait and steadier.

2. a. 14 b. 17 − 10 = 7 c. 14 d. 22 − 9 = 13. Same center; Plot A is more consistent.

3. a. Month 1 b. Month 1

4. Not right. E has the higher center (72 > 68), but F is more consistent (IQR 7 < 28).

5. a. 19 b. 12 c. 19 d. 6. e. Same median; Fountain 2 is more consistent (smaller IQR).

6. Team 1. Its IQR is much smaller (6 vs. 25), so its scores stay close to its median.

7. Store R usually sells more (median 38 vs. 33), but Store S is more consistent (IQR 5 vs. 14).

8. a. Method 1 b. Method 2 (IQR 5 vs. 20) c. Model: Method 2; its typical score is only 4 points lower, and almost every student scores close to it.

9. Median measures center and IQR measures spread; one does not decide the other. Example: Set E has median 60, IQR 30 and Set F has median 55, IQR 8. Compare each measure separately.

10. **A** (7)

11. **A** (Group 1)

Self-check word **MEDIANS**

SHEET 1 · MEDIAN AND OUTLIERS

1. a. median 11 (outlier 48) b. median 72 (outlier 3) c. median 30.5 (outlier 95) d. median 44 (outlier 150) e. median 8 (outlier 34) f. median 54.5 (outlier 9)

2. a. 85 b. 25 min c. 8 students d. The median (25 min); the outlier 85 pulls the mean up, so 8 of 10 students are below it.

3. a. 61 (ordered: 4, 58, 59, 61, 63, 64, 66) b. (61 + 63) ÷ 2 = 62. The median changed by only 1.

4. a. $140 ÷ 7 = $20 b. $14 c. the median; $60 is an outlier, and the mean ($20) is greater than 6 of the 7 amounts

5. No. The median is 36, which sits right in the middle of the main group. Without 210 the median would be (34 + 36) ÷ 2 = 35, so the outlier changes the median by only 1. Here the median changes by only 1; the mean changes much more.

SHEET 2 · COMPARING DISTRIBUTIONS

1. a. A: 25 min, B: 35 min b. A: 30 − 20 = 10 min, B: 45 − 25 = 20 min c. Class B (greater median) d. Class A (smaller IQR)

2. a. Route 3: median 6, IQR 9 − 3 = 6; Route 8: median 6, IQR 14 − 1 = 13 b. Route 3: same median, but a smaller IQR (6 vs 13), so its times are more consistent.

3. a. Sun: median (15 + 16) ÷ 2 = 15.5 cm; Q1 = 12.5, Q3 = 18.5, IQR = 6 cm b. Shade: median (8 + 9) ÷ 2 = 8.5 cm; Q1 = 7.5, Q3 = 10, IQR = 2.5 cm. Sun plants are usually taller; shade plants are more consistent.

4. the same spread (IQR 8 min for both); Shop K's typical delivery is 8 minutes faster (median 28 vs 36).

5. Set E does have the higher center (70 > 64), but Set F is more consistent because its IQR is smaller (6 < 25). A greater median does not mean less spread.

6. **A** (Group 1)

7. **A** (7)

October 2026 conceptual additions — teacher key

Session 1 · Median and Outliers

A: mean 6, median 6. B: mean 10, median 6. The median did not change; the mean increased by 4. For a central individual value, the median is more representative here, while the mean still summarizes total divided by count.

Extension: For fair sharing or total workload: five groups each taking the mean amount 10 would account for the total 50.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Comparing Distributions

A has the shorter typical wait by median, 3 minutes less. B has the narrower middle-half spread, by 6 minutes. Either choice can be justified by the stated priority; neither summary guarantees an individual wait.

Extension: Maximum or upper-tail information and the actual context would help; medians and IQRs alone cannot guarantee a maximum wait.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.